STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Lecture Plan and Review

Today: Section 6.4.

Wednesday: Chapter 1, Sections 1.1, 1.2, and 1.3.

Review of Section 6.3

The Section 6.3 notation emphasized the following numerical measures:

Center	$\bar{x}$ (sample mean), $\widetilde{x}$ (sample median), mode, μ (population mean)
Variation	range, IQR, s (sample standard deviation), σ (population standard deviation)

4 Section 6.4 — Relative Standing, Percentiles, Quartiles, Outliers, and Boxplots

4.1 I. Measures of Relative Standing: z-Scores

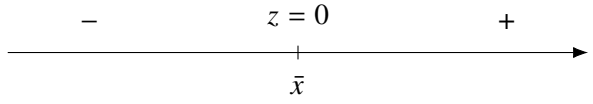
For a sample observation x_i from a dataset with sample mean $\bar{x}$ and sample standard deviation s, the z-score is

$$\boxed{z_i = \frac{x_i - \bar{x}}{s}}.$$

Equivalently,

$$\boxed{x_i = \bar{x} + z_i s}.$$

The z-score tells how many standard deviations a value lies above or below the mean. A positive z-score is above the mean, a negative z-score is below the mean, and $z = 0$ occurs at $\bar{x}$.



The 3-Standard-Deviation Rule

As a rule of thumb, almost all values in a dataset lie within three standard deviations of the mean. In z-score form,

$$\boxed{-3 < z_i < 3}.$$

4.2 II. Percentiles

If the 15th percentile of a dataset is 6.12, then 15% of the ordered dataset is at or below 6.12.

4.3 III. Quartiles

The second quartile is the median of the entire ordered dataset:

$$\boxed{Q_2 = \widetilde{x}}.$$

The first (lower) quartile Q_1 is the median of the lower half, and the third (upper) quartile Q_3 is the median of the upper half.

Even and Odd Sample Sizes

If n is even, the lower half consists of the smallest 50% of the values and the upper half consists of the largest 50%.

If n is odd, Q_2 is the middle value. For the convention used in class, the median is included in both halves: the lower half ends with Q_2, and the upper half begins with Q_2.

Examples: Finding the Three Quartiles

1) For the data $5, 6, 10, 4, 17$, first order the values:

$$4, 5, 6, 10, 17.$$

Thus

$$\boxed{Q_2 = 6, \qquad Q_1 = 5, \qquad Q_3 = 10}.$$

2) For the ordered data

$$1, 8, 10, 17, 21, 24,$$

we have

$$Q_2 = \frac{10 + 17}{2} = 13.5, \qquad \boxed{Q_1 = 8, \quad Q_3 = 21}.$$

3) For the ordered data

$$5, 6, 10, 17, 18, 22, 31,$$

$Q_2 = 17$. Including the median in both halves gives

$$\underbrace{5, 6, 10, 17}_{\text{lower half}} \qquad \underbrace{17, 18, 22, 31}_{\text{upper half}},$$

so

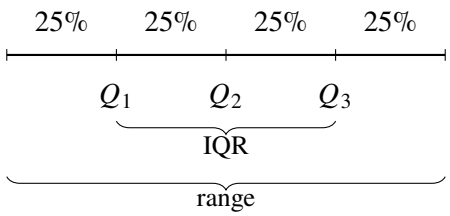
$$\boxed{Q_1 = 8, \qquad Q_2 = 17, \qquad Q_3 = 20}.$$

4.4 IV. Interquartile Range

The *interquartile range* is

$$\boxed{\text{IQR} = Q_3 - Q_1}.$$

It measures the spread of the middle 50% of an ordered dataset.

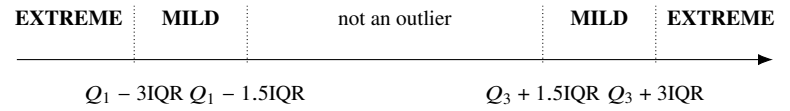


4.5 V. Outliers

A value is an *outlier* if it is at least 1.5(IQR) away from the nearest quartile. If it is at least 3(IQR) away from the nearest quartile, it is an *extreme outlier*; otherwise it is a *mild outlier*.

The four reference points are

$$Q_1 - 3\text{IQR}, \qquad Q_1 - 1.5\text{IQR}, \qquad Q_3 + 1.5\text{IQR}, \qquad Q_3 + 3\text{IQR}.$$



In a boxplot, a mild outlier is indicated by a circle $\circ$, while an extreme outlier is indicated by an asterisk $*$. Most datasets do not have outliers.

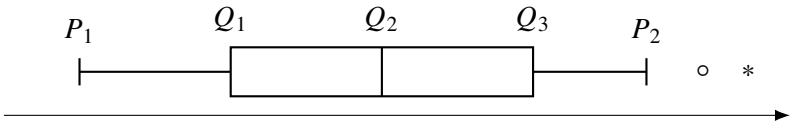
4.6 VI. Boxplots

A boxplot is also called a *box-and-whiskers plot*. To construct a boxplot:

1. Order the dataset.
2. Find Q_2, Q_1, Q_3, and IQR.
3. Determine whether there are outliers.
4. Find

   $$P_1 = \text{smallest value that is not an outlier,} \qquad \text{and} \qquad P_2 = \text{largest value that is not an outlier.}$$

5. Draw the boxplot.



The Five-Number Summary

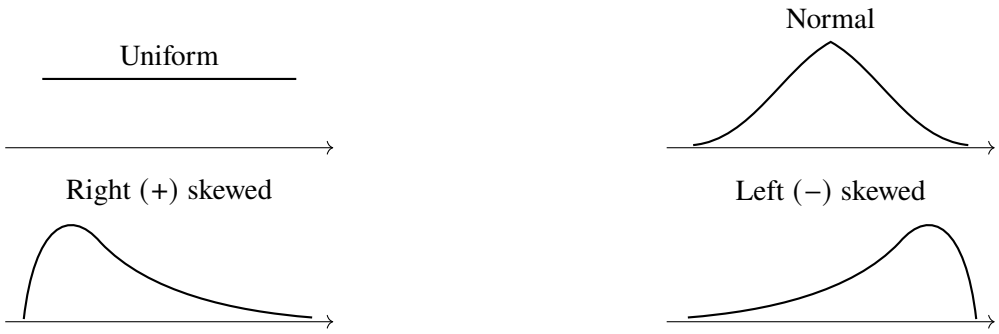
The five-number summary is

$$\boxed{\{\text{smallest value, } Q_1, Q_2, Q_3, \text{ largest value}\}}.$$

The smallest and largest values in the five-number summary are the actual dataset extremes, even when one of them is plotted separately as an outlier in the boxplot.

Four Schematic Distribution Shapes

The four cases considered were uniform, normal, right (+) skewed, and left (−) skewed.



Example 6.4.1 — Site II

(a) Boxplot, showing the steps

1. Order the dataset:

$$1, 9, 10, 12, 13, 14, 15, 15, 17, 18, 19, 20, 20, 21, 22, 23, 24, 25.$$

2. Quartiles and IQR:

$$Q_2 = \frac{17 + 18}{2} = 17.5, \qquad Q_1 = 13, \qquad Q_3 = 21,$$

$$\boxed{\text{IQR} = Q_3 - Q_1 = 21 - 13 = 8}.$$

3. Outliers:

$$Q_1 - 1.5\text{IQR} = 13 - 12 = 1, \qquad Q_1 - 3\text{IQR} = 13 - 24 = -11.$$

Thus 1 is a mild outlier on the left. On the right,

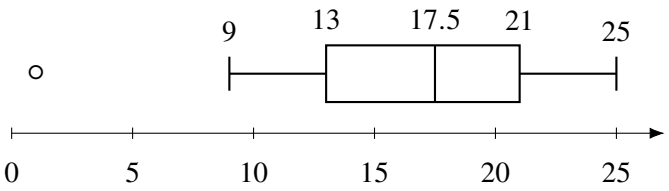
$$Q_3 + 1.5\text{IQR} = 21 + 12 = 33,$$

so there is no outlier on the right.

4. Whisker endpoints:

$$\boxed{P_1 = 9, \qquad P_2 = 25}.$$

5. Boxplot:



(b) Five-Number Summary

$$\boxed{\{1, 13, 17.5, 21, 25\}}.$$

(c) Shape

The distribution is left-skewed.

Example 6.4.1 — Site I

(a) Stem-and-Leaf Plot Using Two Lines per Stem

Note from class: The stem-and-leaf plot on p. 207 has an error. Confirm its ordering and scale/key with Venema.

The stem gives the tens digit and the leaf gives the ones digit. Each stem appears on two rows: the first contains leaves 0–4, and the second contains leaves 5–9.

Stem	Leaves
1	0 2 2 3 3 3 4
1	6 6 7 8 9
2	0 1 4
2	5 7
3	1 2 2
3	6 7
4	
4	9

For example, 3 | 2 represents 32. The sample size is

$$\boxed{n = 23}.$$

(b) Quartiles, Outliers, and Boxplot

The ordered data are

$$10, 12, 12, 13, 13, 13, 14, 16, 16, 17, 18, 19, 20, 21, 24, 25, 27, 31, 32, 32, 36, 37, 49.$$

Using the odd-n convention from class,

$$\boxed{Q_2 = 19, \qquad Q_1 = 13.5, \qquad Q_3 = 29},$$

and

$$\boxed{\text{IQR} = 29 - 13.5 = 15.5}.$$

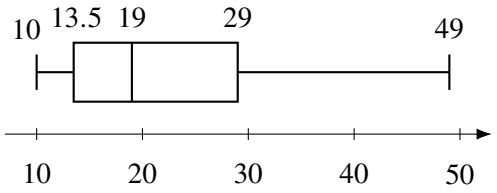
The mild-outlier fences are

$$Q_1 - 1.5\text{IQR} = 13.5 - 23.25 = -9.75 < 0,$$

$$Q_3 + 1.5\text{IQR} = 29 + 23.25 = 52.25.$$

There are no outliers on either side, so

$$\boxed{P_1 = 10, \qquad P_2 = 49}.$$



The five-number summary is

$$\boxed{\{10, 13.5, 19, 29, 49\}},$$

and the distribution is positively skewed (right-skewed).

Example 6.4.2

The dataset was organized by a stem-and-leaf plot with $n = 21$:

Stem	Leaves
0	8
1	2
2	0 7
3	0 2 5 6
4	0 0 0 0 1 2 5 7
5	0 2
6	1
7	
8	9
9	
10	8

Here again, the stem gives tens and the leaf gives ones.

Quartiles and IQR

Using the class convention for odd n,

$$\boxed{Q_2 = 40, \qquad Q_1 = 32, \qquad Q_3 = 47},$$

so

$$\boxed{\text{IQR} = 47 - 32 = 15}.$$

Outliers

On the left,

$$Q_1 - 1.5\text{IQR} = 32 - 22.5 = 9.5, \qquad Q_1 - 3\text{IQR} = 32 - 45 = -13.$$

Thus 8 is a mild outlier.

On the right,

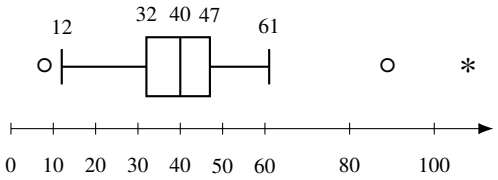
$$Q_3 + 1.5\text{IQR} = 47 + 22.5 = 69.5, \qquad Q_3 + 3\text{IQR} = 47 + 45 = 92.$$

Therefore,

61 is not an outlier, 89 is a mild outlier, 108 is an extreme outlier.

The whisker endpoints are

$$\boxed{P_1 = 12, \qquad P_2 = 61}.$$



The five-number summary is

$$\boxed{\{8, 32, 40, 47, 108\}}.$$

Homework and Next Class

Homework: Problems 28, 32, and 33, pp. 219–223.

Wednesday: Chapter 1, Sections 1.1, 1.2, and 1.3.